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DEPARTEMENT OF COMPUTER SCIENCE AND ENGINEERING

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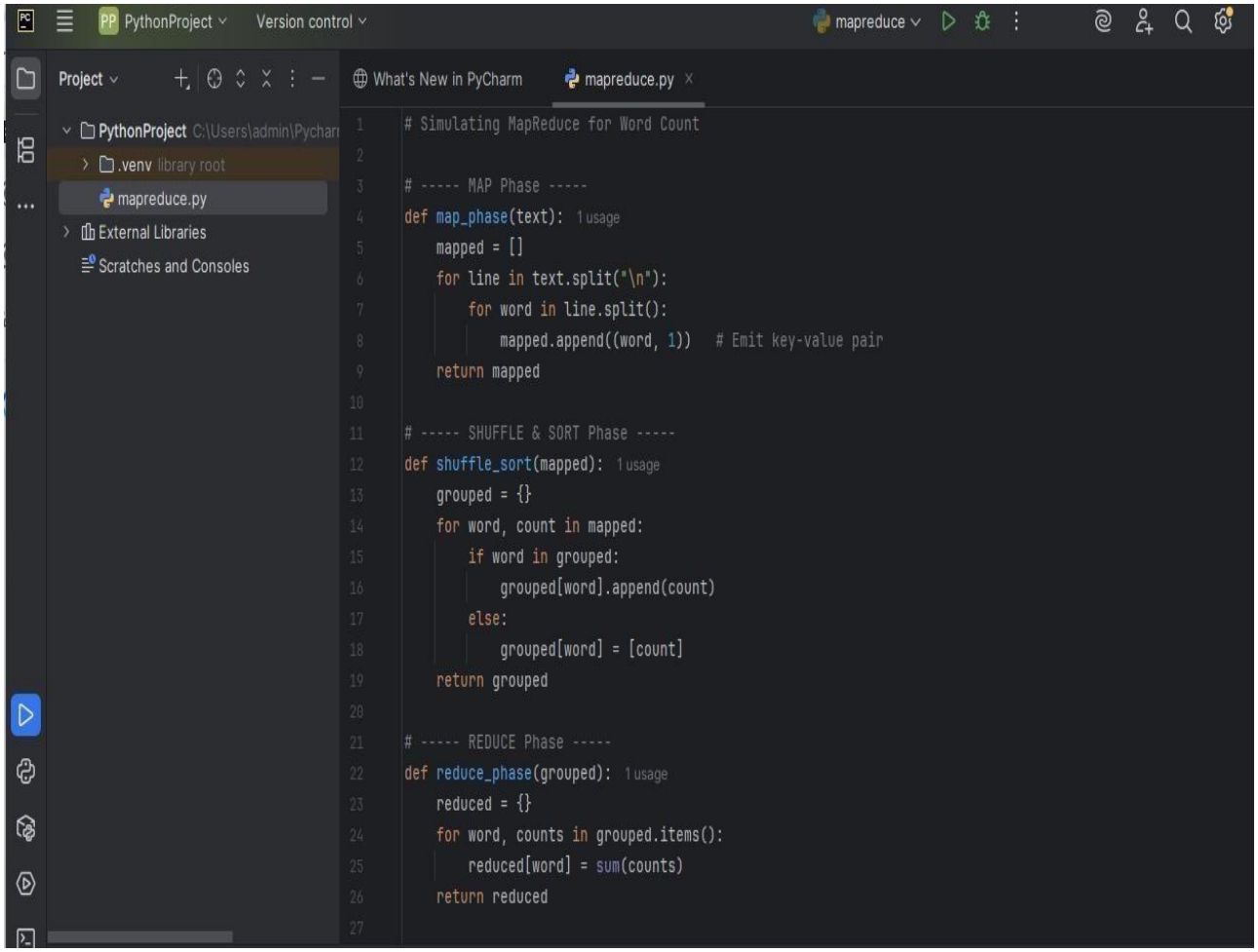
Experiment 7

Title: Demonstrate the Use of Map and Reduce Tasks.




```python
# Simulating MapReduce for Word Count

# ----- MAP Phase -----
def map_phase(text):
    mapped = []
    for line in text.split("\n"):
        for word in line.split():
            mapped.append((word, 1))   # Emit key-value pair
    return mapped

# ----- SHUFFLE & SORT Phase -----
def shuffle_sort(mapped):
    grouped = {}
    for word, count in mapped:
        if word in grouped:
            grouped[word].append(count)
        else:
            grouped[word] = [count]
    return grouped

# ----- REDUCE Phase -----
def reduce_phase(grouped):
    reduced = {}
    for word, counts in grouped.items():
        reduced[word] = sum(counts)
    return reduced
```

```python
    return reduced


# Sample input (like from data.txt)
input_text = """cat dog cat lion
dog tiger cat
lion tiger dog"""

print("Input Data:\n", input_text)

# Run MapReduce simulation
mapped_output = map_phase(input_text)
print("\n1 MAP Phase Output:")
print(mapped_output)

grouped_data = shuffle_sort(mapped_output)
print("\n2 SHUFFLE & SORT Phase:")
print(grouped_data)

final_output = reduce_phase(grouped_data)
print("\n3 REDUCE Phase Output (Final Result):")
print(final_output)
```

```
C:\Users\admin\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\admin\PycharmProjects\PythonProject\mapreduce.py
Input Data:
 cat dog cat lion
dog tiger cat
lion tiger dog

1 MAP Phase Output:
[('cat', 1), ('dog', 1), ('cat', 1), ('lion', 1), ('dog', 1), ('tiger', 1), ('cat', 1), ('lion', 1), ('tiger', 1), ('dog', 1)]

2 SHUFFLE & SORT Phase:
{'cat': [1, 1, 1], 'dog': [1, 1, 1], 'lion': [1, 1], 'tiger': [1, 1]}

3 REDUCE Phase Output (Final Result):
{'cat': 3, 'dog': 3, 'lion': 2, 'tiger': 2}

Process finished with exit code 0
```